

Richy Koshy

Python Engineer — GenAI, Backend & Automation

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REFERENCE Corban Wells, VP of Technology Product Management, PRMI — *“Conscientious ownership, earnest curiosity, and diligent follow-through.”* Full recommendation letter available on request.

Python engineer with 4+ years shipping production GenAI, backend, and automation systems for a US mortgage originator — from requirements to production: an enterprise Azure RAG platform, a custom Encompass REST API SDK, and 13 production automations.

WORK EXPERIENCE

Software Engineer — Antz AI Solutions

Client: Primary Residential Mortgage, Inc. (PRMI), US — sole engineer owning requirements, architecture, build, UAT, and production.

- **MAX AI — enterprise RAG platform.** Python backend across three Azure services: FastAPI chat with NDJSON streaming, Durable Functions ingestion pipeline, and utility APIs. Hybrid retrieval on Azure AI Search (vector + semantic ranking, HyDE query expansion), Azure OpenAI (GPT-4o, ada-002), Azure AD auth with multi-tenant access mapping in SQL Server; Dockerized ingestion via Azure Container Registry. **PRMI's primary knowledge-retrieval tool organization-wide.**
- **Final Docs refactor.** Rebuilt a monolithic automation into config-driven dynamic dispatch (YAML → handlers): 4 new investors onboarded in 1 month vs. a full quarter before; onboarding cut from 2–4 weeks to days; adopted as the template for 2 other initiatives.

“PRMI now treats investor expansion as a business decision, not a dev-capacity constraint.” — Corban Wells, VP Technology Product Management, PRMI
- **Encompass REST API SDK.** Python SDK wrapping Encompass Cloud APIs: OAuth2 with token caching, concurrent paginated pipeline queries, safe loan updates, eFolder document CRUD, disclosure tracking.
- **13 production RPA automations** (Power Automate + Python) across ECOA adverse action, TRID, HMDA, MERS sync, final docs, HFA exceptions, Blend credit management, warehouse collateral, and suspense resolution across 9 investors — plus 3 OCR pipelines (Tesseract + Azure Form Recognizer) extracting credit data from PDF reports.
- **Delivery ownership.** Led BRD/PRD gathering, flow diagrams, first-call sign-offs, UAT coordination, and documentation for every delivered automation.
- **HydroQ API** (client: TechSpan Engineering, IoT). FastAPI + SQLAlchemy REST API for remote water-quality monitoring with a generic CRUD layer over PostgreSQL and MSSQL; replaced 2–4 hours of manual onboarding per site with API-driven workflows; full endpoint test coverage.

Industrial Automation Engineer — Goodyear South Asia Tyres

- Programmed and maintained PLCs and variable-frequency drives across tyre-curing and material-handling lines.
- Led breakdown troubleshooting on production-critical electrical systems, minimizing line downtime.
- Commissioned CTMS & BMS systems alongside Rockwell and Honeywell engineers.

SKILLS

GENAI / RAG

Azure OpenAI (GPT-4o, ada-002), Azure AI Search, hybrid vector + semantic retrieval, HyDE, ingestion pipelines

BACKEND

Python, FastAPI, asyncio, SQLAlchemy, Pydantic, REST API design, OAuth2, pytest

CLOUD & INFRA

Azure (Durable Functions, App Services, Container Registry, Blob/Table Storage), Docker, SQL Server, PostgreSQL

AUTOMATION

Power Automate, RPA, Encompass REST API, Blend API, Tesseract / Azure Form Recognizer OCR, Pandas

GITHUB — GITHUB.COM/RICHIE-RK

rag-app-azure — enterprise Azure RAG app: hybrid search (vector + semantic + BM25), NDJSON streaming, MSAL SSO, FastAPI + Azure Functions + React

Bible-VerseFinder-AI — AI-powered Bible verse search engine with LLM summarization and full chapter reading

PyTeslaCoil — pure Python implementation of Java TeslaCoil

EncompassRest-RAG-Assistant — natural-language RAG over ICE Mortgage Encompass API docs, hybrid retrieval, multi-LLM

azure-search-chunk-viewer — Python web app for browsing and analyzing Azure AI Search index chunks

EDUCATION & AWARDS

BE, Electrical Engineering

University of Mumbai

Best Paper, Intl. Conference on Advanced Trends in Engineering

1st Place, Tantragyan National Project Competition